

OS ASSIGNMENT 1.B

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Div & batch: B / B3

PS:Write a shell script that implements various file and directory related commands.

Code:

```
#!/bin/bash
```

```
# =====  
#   LINUX FILE MANAGEMENT SYSTEM  
# =====
```

```
while true  
do
```

```
    clear
```

```
    echo "===== "  
    echo "   LINUX FILE MANAGEMENT SYSTEM"  
    echo "===== "  
    echo  
    echo "1. Create a File"  
    echo "2. Copy a File"  
    echo "3. Move a File"  
    echo "4. Delete a File"  
    echo "5. Rename a File"  
    echo "6. Display File Content"  
    echo "7. Write into a File"  
    echo "8. Append Text to a File"  
    echo "9. Display File Size"  
    echo "10. Compare Two Files"  
    echo "11. Combine Two Files"  
    echo "12. Count Lines, Words and Characters"  
    echo "13. Create a Directory"  
    echo "14. Delete a Directory"  
    echo "15. List Directory Contents"  
    echo "16. Search for a File"  
    echo "17. Show Current Directory"  
    echo "18. Change Directory"  
    echo "19. Create Empty File (Touch)"
```

```
echo "20. Exit"
echo
echo "=====
```

```
read -p "Enter your choice: " choice
```

```
clear
```

```
case $choice in
```

```
# -----
# 1. CREATE FILE
# -----
```

```
1)
```

```
echo "----- CREATE FILE -----"
echo
```

```
read -p "Enter file name: " filename
```

```
if [ -e "$filename" ]
then
```

```
    echo "File already exists."
```

```
else
```

```
    touch "$filename"
```

```
    if [ $? -eq 0 ]
```

```
    then
```

```
        echo "File created successfully."
```

```
    else
```

```
        echo "Error creating file."
```

```
    fi
```

```
fi
```

```
::
```

```
# -----
# 2. COPY FILE
# -----
```

```
2)
```

```
echo "----- COPY FILE -----"
echo
```

```
read -p "Enter source file: " source
read -p "Enter destination: " destination
```

```
if [ -f "$source" ]
then
    cp "$source" "$destination"

    if [ $? -eq 0 ]
    then
        echo "File copied successfully."
    else
        echo "Error copying file."
    fi
else
    echo "Source file does not exist."
fi
;;
```

```
# -----
# 3. MOVE FILE
# -----
```

```
3)
echo "----- MOVE FILE -----"
echo

read -p "Enter file name: " source
read -p "Enter destination: " destination

if [ -e "$source" ]
then
    mv "$source" "$destination"

    if [ $? -eq 0 ]
    then
        echo "File moved successfully."
    else
        echo "Error moving file."
    fi
else
    echo "File does not exist."
fi
;;
```

```
# -----  
# 4. DELETE FILE  
# -----
```

```
4)  
echo "----- DELETE FILE -----"  
echo  
  
read -p "Enter file name: " filename  
  
if [ -f "$filename" ]  
then  
    read -p "Are you sure you want to delete it? (y/n): " answer  
  
    if [ "answer" = "y" ] || [ "answer" = "Y" ]  
    then  
        rm "$filename"  
  
        if [ $? -eq 0 ]  
        then  
            echo "File deleted successfully."  
        else  
            echo "Error deleting file."  
        fi  
    else  
        echo "Delete operation cancelled."  
    fi  
else  
    echo "File does not exist."  
fi  
;;
```

```
# -----  
# 5. RENAME FILE  
# -----
```

```
5)  
echo "----- RENAME FILE -----"  
echo  
  
read -p "Enter old file name: " oldname
```

```
read -p "Enter new file name: " newname
```

```
if [ -e "$oldname" ]
```

```
then
```

```
    mv "oldname" "newname"
```

```
    if [ $? -eq 0 ]
```

```
    then
```

```
        echo "File renamed successfully."
```

```
    else
```

```
        echo "Error renaming file."
```

```
    fi
```

```
else
```

```
    echo "File does not exist."
```

```
fi
```

```
::
```

```
# -----
```

```
# 6. DISPLAY FILE CONTENT
```

```
# -----
```

```
6)
```

```
echo "----- DISPLAY FILE CONTENT ----"
```

```
echo
```

```
read -p "Enter file name: " filename
```

```
if [ -f "$filename" ]
```

```
then
```

```
    echo
```

```
    echo "----- FILE CONTENT ----"
```

```
    cat "$filename"
```

```
    echo
```

```
    echo "-----"
```

```
else
```

```
    echo "File does not exist."
```

```
fi
```

```
::
```

```
# -----
```

```
# 7. WRITE INTO FILE
```

```
# -----
```

```
7)
```

```
echo "----- WRITE INTO FILE -----"
```

```
echo
```

```
read -p "Enter file name: " filename
```

```
read -p "Enter text: " text
```

```
echo "text" > "filename"
```

```
if [ $? -eq 0 ]
```

```
then
```

```
    echo "Text written successfully."
```

```
else
```

```
    echo "Error writing to file."
```

```
fi
```

```
::
```

```
;;
```

```
# -----
```

```
# 8. APPEND TEXT TO FILE
```

```
# -----
```

```
8)
```

```
echo "----- APPEND TEXT TO FILE -----"
```

```
echo
```

```
read -p "Enter file name: " filename
```

```
read -p "Enter text: " text
```

```
echo "text" >> "filename"
```

```
if [ $? -eq 0 ]
```

```
then
```

```
    echo "Text added successfully."
```

```
else
```

```
    echo "Error appending text."
```

```
fi
```

```
::
```

```
;;
```

```
# -----
```

9. DISPLAY FILE SIZE

9)

```
echo "----- FILE SIZE -----"
```

```
echo
```

```
read -p "Enter file name: " filename
```

```
if [ -e "$filename" ]
```

```
then
```

```
    ls -lh "$filename"
```

```
else
```

```
    echo "File does not exist."
```

```
fi
```

```
::
```

10. COMPARE TWO FILES

10)

```
echo "----- COMPARE TWO FILES -----"
```

```
echo
```

```
read -p "Enter first file: " file1
```

```
read -p "Enter second file: " file2
```

```
if [ -f "$file1" ] && [ -f "$file2" ]
```

```
then
```

```
    cmp -s "$file1" "$file2"
```

```
    if [ $? -eq 0 ]
```

```
    then
```

```
        echo "Both files are identical."
```

```
    else
```

```
        echo "Files are different."
```

```
    fi
```

```
else
```

```
    echo "One or both files do not exist."
```

```
fi
```

```
::
```

```
# -----  
# 11. COMBINE TWO FILES  
# -----
```

```
11)  
  echo "----- COMBINE TWO FILES -----"  
  echo  
  
  read -p "Enter first file: " file1  
  read -p "Enter second file: " file2  
  read -p "Enter output file: " output  
  
  if [ -f "file1" ] && [ -f "file2" ]  
  then  
    cat "file1" "file2" > "$output"  
  
    if [ $? -eq 0 ]  
    then  
      echo "Files combined successfully."  
      echo "Output file: $output"  
    else  
      echo "Error combining files."  
    fi  
  else  
    echo "One or both files do not exist."  
  fi  
  ;;
```

```
# -----  
# 12. COUNT LINES, WORDS AND CHARACTERS  
# -----
```

```
12)  
  echo "----- FILE COUNT -----"  
  echo  
  
  read -p "Enter file name: " filename  
  
  if [ -f "$filename" ]  
  then  
    echo  
    echo "Lines Words Characters"
```

```
        wc "$filename"
    else
        echo "File does not exist."
    fi
;;
```

```
# -----
# 13. CREATE DIRECTORY
# -----
```

```
13)
    echo "----- CREATE DIRECTORY -----"
    echo

    read -p "Enter directory name: " dirname

    if [ -d "$dirname" ]
    then
        echo "Directory already exists."
    else
        mkdir "$dirname"

        if [ $? -eq 0 ]
        then
            echo "Directory created successfully."
        else
            echo "Error creating directory."
        fi
    fi
;;
```

```
# -----
# 14. DELETE DIRECTORY
# -----
```

```
14)
    echo "----- DELETE DIRECTORY -----"
    echo

    read -p "Enter directory name: " dirname

    if [ -d "$dirname" ]
```

```

then
    rmdir "$dirname"

    if [ $? -eq 0 ]
    then
        echo "Directory deleted successfully."
    else
        echo "Directory is not empty."
        echo "Only empty directories can be deleted using rmdir."
    fi
else
    echo "Directory does not exist."
fi
;;

```

```

# -----
# 15. LIST DIRECTORY CONTENTS
# -----

```

```

15)
echo "----- DIRECTORY CONTENTS -----"
echo

read -p "Enter directory path (use . for current directory): " dirname

if [ -d "$dirname" ]
then
    ls -la "$dirname"
else
    echo "Directory does not exist."
fi
;;

```

```

# -----
# 16. SEARCH FOR A FILE
# -----

```

```

16)
echo "----- SEARCH FILE -----"
echo

read -p "Enter directory to search: " dirname

```

```

read -p "Enter file name to search: " filename

if [ -d "$dirname" ]
then
    result=(find"dirname" -name "$filename" 2>/dev/null)

    if [ -n "$result" ]
    then
        echo
        echo "File found:"
        echo "$result"
    else
        echo "File not found."
    fi
else
    echo "Directory does not exist."
fi
;;

```

```

# -----
# 17. SHOW CURRENT DIRECTORY
# -----

```

```

17)
echo "----- CURRENT DIRECTORY -----"
echo

echo "You are currently in:"
pwd
;;

```

```

# -----
# 18. CHANGE DIRECTORY
# -----

```

```

18)
echo "----- CHANGE DIRECTORY -----"
echo

read -p "Enter directory path: " dirname

if [ -d "$dirname" ]

```

```

then
    cd "$dirname"

    if [ $? -eq 0 ]
    then
        echo "Directory changed successfully."
        echo
        echo "Current directory:"
        pwd
    else
        echo "Unable to change directory."
    fi
else
    echo "Directory does not exist."
fi
;;

```

```

# -----
# 19. TOUCH FILE
# -----

```

```

19)
    echo "----- TOUCH FILE -----"
    echo

    read -p "Enter file name: " filename

    touch "$filename"

    if [ $? -eq 0 ]
    then
        echo "File created/updated successfully."
    else
        echo "Error performing touch operation."
    fi
    ;;

```

```

# -----
# 20. EXIT
# -----

```

20)

```

echo
echo "=====
echo "  Thank you for using File Management System"
echo "=====
exit 0
;;

```

```

# -----
# INVALID CHOICE
# -----

```

```

*)
echo
echo "Invalid choice!"
echo "Please enter a number from 1 to 20."
;;

```

```

esac

```

```

echo
read -p "Press Enter to continue..."

```

```

done

```

Output:

```

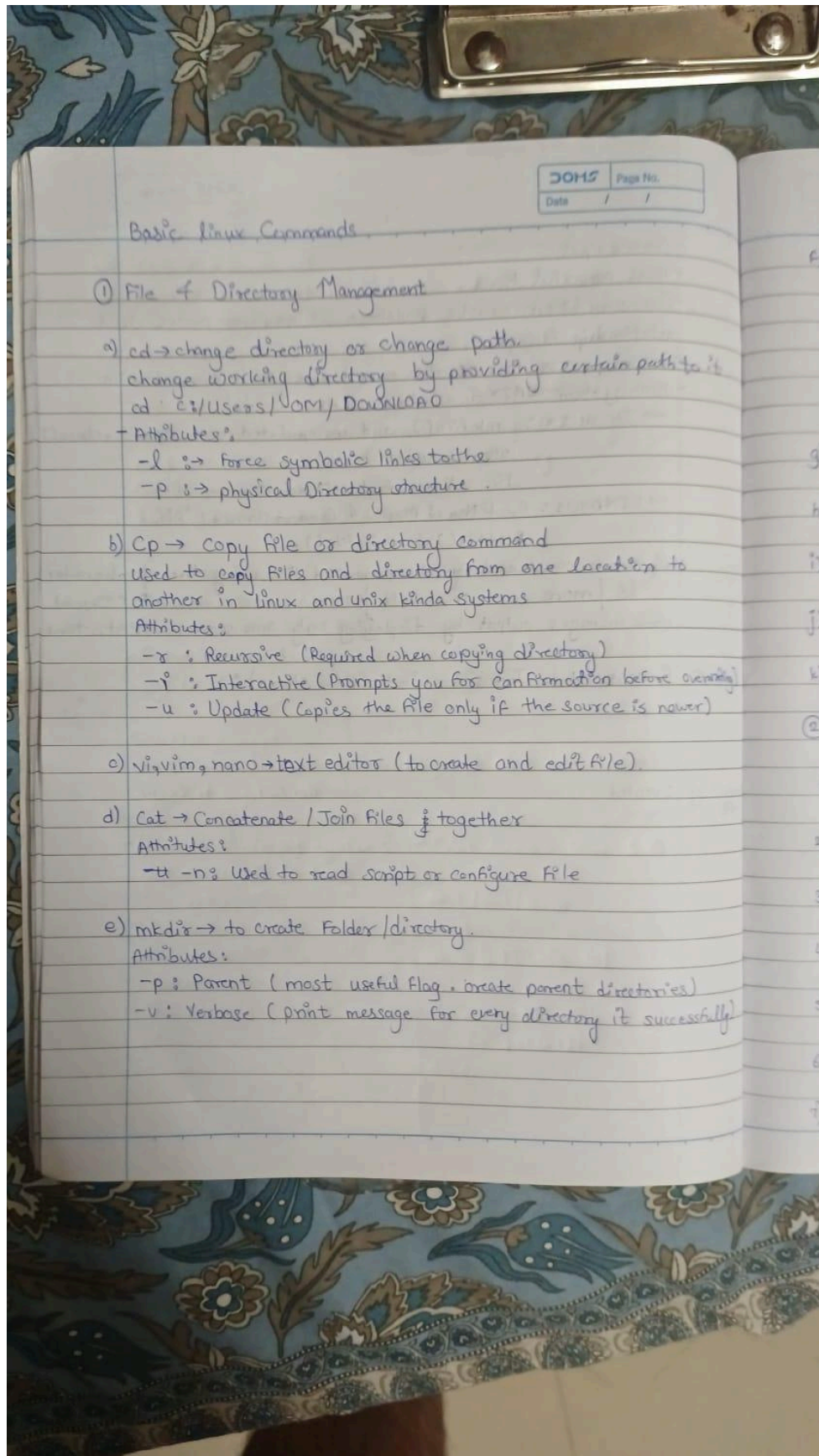
omion@OM: ~
=====
          LINUX FILE MANAGEMENT SYSTEM
=====

1.  Create a File
2.  Copy a File
3.  Move a File
4.  Delete a File
5.  Rename a File
6.  Display File Content
7.  Write into a File
8.  Append Text to a File
9.  Display File Size
10. Compare Two Files
11. Combine Two Files
12. Count Lines, Words and Characters
13. Create a Directory
14. Delete a Directory
15. List Directory Contents
16. Search for a File
17. Show Current Directory
18. Change Directory
19. Create Empty File (Touch)
20. Exit

=====
Enter your choice: 2|

```

Notebook:



Basic Linux Commands

① File & Directory Management

a) `cd` → change directory or change path.
change working directory by providing certain path to it.
`cd : C:/Users/COM/Downloads`

Attributes:

- `-l` : force symbolic links to the
- `-p` : physical Directory structure

b) `cp` → copy file or directory command
used to copy files and directory from one location to another in linux and unix kind systems

Attributes:

- `-r` : Recursive (Required when copying directory)
- `-i` : Interactive (Prompts you for confirmation before overwriting)
- `-u` : Update (Copies the file only if the source is newer)

c) `vi`, `vim`, `nano` → text editor (to create and edit file).

d) `Cat` → Concatenate / Join files together

Attributes:

- `-t` - `-n` : Used to read script or configure file

e) `mkdir` → to create folder / directory.

Attributes:

- `-p` : Parent (most useful flag, create parent directories)
- `-v` : Verbose (print message for every directory it successfully)

practical assignments : Program
Notebook Jancopy.
program output.

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LAB

Switch case program.

ls commands : cp , rm , mkdir : attributes (Advance).

1. del directory.	(9)	rm filename.txt
2. mkdir.	(8)	echo -n " " > f-name.txt
3. show content of directory.	(4)	
4. creating directory.	(1)	touch "\$file"
5. moving directory.	(12)	
6. copying directory.	(11)	
7. show directory X		
8. Rename directory	(6)	
9. Write content in directory	(5)	
10. check file exist	(2)	
11. show file permissions	(3)	ls -l
12. change file permission	(7)	
13. count lines	(13)	
14. count words	(14)	
15. count characters	(15)	
16. List files	(10)	
17. Exit.		

Switch case syntax in shell:

a variable choice (ch)

case \$ch in

1) echo "cmd"

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

...

esac

echo -n "invalid ch"

done

if [-e "\$f_name"] then

echo "ls -l"

exit (exit - failure) cmd.

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Theory